

Data Science

CMPT 353 D100

Individual Project Report

CMPT 353 Final Project: Social Media Analysis

Analyzing Social Media Content to Understand Engagement Patterns

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<https://github.sfu.ca/gkb8/CMPT353-Project>

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Overview

For my final project in CMPT 353, I chose to analyze my personal Instagram content especially reels and posts to explore the following research question:

“Which core engagement type has the strongest correlation with high views?”

As someone who has gained over 15,000 followers in just three months, I’ve seen firsthand how unpredictable social media growth can be. People often give advice like “just get more likes” or try to “boost your comments and shares,” but I wanted to see which engagement type had the highest correlation with high views. My focus was on the core engagement areas of likes, comments, shares, and shares, and whether higher numbers in these metrics actually translate into more views. This is important because brands are far more likely to with creators whose content consistently reaches a large audience. Knowing which engagement type is most likely strongly linked to higher views can help guide both my own strategy and the way other creators focus their efforts.

As a fitness content creator, I frequently share workouts, progress updates, recipes, and motivation posts on Instagram. While some posts go viral, others underperform despite similar style and effort. This project allowed me to take a data-driven look at my own content to identify which engagement types truly matter for driving reach.

Data Used

To investigate this question, I downloaded my personal Instagram data export, which included insights for both reels and regular posts. The files “reels.json” and “posts.json” contained key engagement metrics such as likes, comments, shares, shares, plays (views), and timestamps

From each post, I extracted and manually engineered several new features to use for analysis, including:

- **Word count and Caption length** (long if ≥ 50 words).
- **Hashtag count**: Total number of hashtags used in the caption.
- **Posting hour and weekday**: To capture potential timing effects.
- **Source type**: Whether the content was a reel or a regular photo post.
- **Trial reel status**: A boolean feature indicating whether the reel was shown to non-followers (i.e., trial reels).
- **Duplicate caption indicator**: Flagged for reused captions.
- **High view threshold**: I defined a post as “high-performing” if it received more than 100,000 views because it usually signals a standout or viral post on my account.

- **Engagement ratios:** Likes per view, comments per view, shares per view, saves per view showed standardized metrics to compare posts of different sizes.

These features represent attributes that are available at the time of posting, which makes them useful for building predictive models that could inform future content decisions.

Data Analyzation Techniques

To investigate which engagement type was most strongly associated with high views, I applied a combination of descriptive statistics, hypothesis testing, and machine learning classification models.

First, I conducted exploratory data analysis (EDA) to better understand the dataset. This involved visualizing audience demographics such as gender, age, and location to see the composition of my followers. I also examined posting patterns, including the distribution of reels versus regular posts, the days and hours I typically post, and how these factors might relate to engagement. In addition, I analyzed the distributions of key engagement ratios, like per view, comments per view, shares per view, and saves per view to identify trends or patterns.

Next, I performed hypothesis testing to compare engagement metrics between high-view posts (high-views if $\geq 100,000$ views) and low-view posts. For each engagement ratio, I ran independent t-tests and Mann-Whitney U tests to determine whether the observed differences between the two groups were statistically significant. This provided a quantitative foundation for understanding which engagement types were most strongly linked to higher view counts.

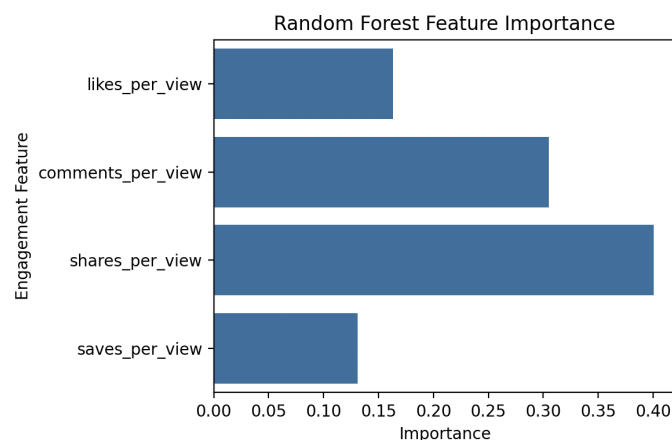
I then applied predictive modeling techniques to classify posts as either high-view or low-view. Two models were used: Logistic Regression, which served as a baseline linear classifier, and Random Forest, which allowed for capturing non-linear relationships. Since high-view posts were relatively rare, I addressed class imbalance by upsampling the minority class in the training data. Both models were evaluated using accuracy, precision, recall, and F-1 score, and feature importance from the Random Forest model was analyzed to identify the most predictive engagement metrics.

Finally, I explored clustering and dimensionality reduction to see if posts naturally grouped based on engagement characteristics. I scaled the features and applied Principal Component Analysis (PCA) to reduce them to two components, capturing over 80% of the variance in the data. K-means clustering with $k=2$ was then used to group posts, and then results were visualized on a PCA scatter plot to assess whether these clusters aligned with high- and low-engagement patterns.

Results and Visualizations

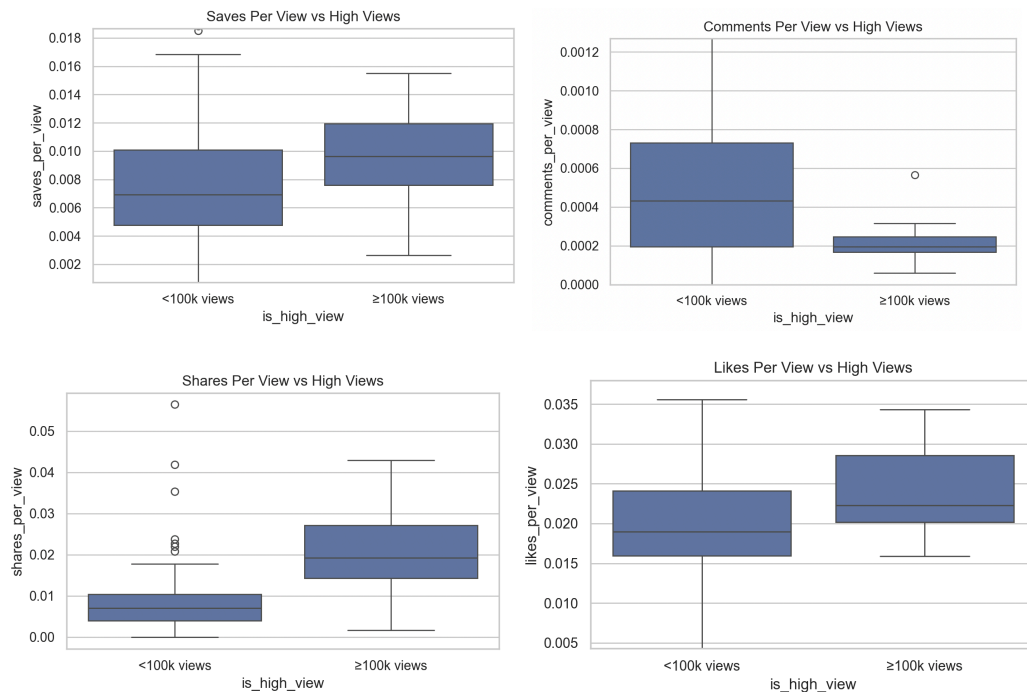
The analysis compared engagement ratios between high-view posts ($\geq 100,000$ views) and low-view posts using both t-tests and Mann-Whitney U tests. Shares per views had the strongest and most statistically significant link to high views (0.0212 vs 0.0082; $t = 4.163$, $p = 0.0009$; $U = 1670$, $p < 0.0001$), suggesting that posts which are shared more frequently relative to views are much more likely to go viral. Likes per view were also higher in the high-view group (0.0257 vs 0.0213) with statistical significance in the Mann-Whitney test ($p = 0.0144$) but not the t-test ($p = 0.1188$), indicating a weaker but still possible relationship. Saves per view showed a smaller but still statistically significant difference in the Mann-Whitney test (0.0101 vs 0.0086; $p = 0.0374$), implying that saved posts might have slightly better longevity or reach. Interestingly, comments per view are significantly higher in the low-view group (0.0007 vs 0.0002; $t = -2.805$, $p = 0.0057$; $U = 551$, $p = 0.0088$), which may indicate that higher engagement through comments does not necessarily translate to broader reach and could be more common on niche or discussion-heavy posts.

To further explore predictive power, two classification models were developed to determine whether a post would be high-view or low-view based on its engagement ratios. Logistic Regression, serving as a baseline linear classifier, achieved 84% accuracy but struggled to identify high-view posts, with a recall score of only 0.17 for this class. In contrast, the Random Forest model achieved 87% accuracy and improved recall for high-view posts to 0.33, indicating that it could capture non-linear relationships between engagement metrics and view count. An analysis of features important within the Random Forest model revealed that shares per view was the strongest predictor. This reinforces the statistical findings that shares per views plays a particularly critical role in driving high view counts.



To better understand engagement differences between high-view and low-view posts, I created boxplots for likes per view, comments per view, saves per view, and shares per view. The plots compare the distribution of these ratios for posts under 100k views versus those with 100k or

more. Shares per view showed the largest difference, in high-view posts having much higher values. Likes and saves per view were slightly higher in high-view posts, while comments per view were higher in low-view posts.



In addition to predictive modeling, clustering techniques were applied to investigate whether posts naturally grouped according to engagement characteristics. Principal Component Analysis (PCA) was first used to reduce the engagement features to two components, capturing a combined 81% of the variance in the data (60% from PC1 and 21% from PC2). K-means clustering with $k=2$ was then performed on the PCA-reduction dataset. The resulting scatter plot below showed some separation between the two clusters suggesting distinct engagement patterns, though there was still substantial overlap. This indicates that while certain engagement behaviours are associated with high or low counts, these factors alone do not perfectly classify posts into distinct performance categories.



Conclusion

The analysis showed that shares per view had the strongest and most consistent link to high views, far outweighing likes, comments, or saves. While likes and saves had smaller positive effects, comments were actually higher on low-view posts, suggesting they don't necessarily drive reach. These results highlight that content people feel compelled to share whether relatable, educational, or emotionally engaging is more likely to go viral.

Limitations

The analysis had several notable limitations. First, the dataset was heavily imbalanced, with only 14 posts exceeding 100k views. This small sample size in the high-view category made it difficult for the models, especially Random Forest and Logistic Regression, to accurately learn patterns for predicting viral posts.

Another limitation was the lack of gender-specific engagement data at the post-level. While overall audience demographics were available, Instagram does not provide female versus male audience breakdowns for individual posts, preventing a deeper analysis into strategies for growing a female audience.

There were also data formatting challenges. In my old CSV export, emojis in my captions were replaced by random symbols. The issue persisted until I realized that the timestamps were not in Pacific Time. Once corrected, the emojis displayed properly, but this experience highlighted how time zone mismatches can interfere with text encoding and parsing.

Lastly, the dataset did not capture newer engagement metrics. For example, Instagram introduced a posting feature just this week while finishing my analysis. This is likely to become a major driver of reach and engagement in the coming months, but its impact could not be measured in the current study.

Project Experience Summary

This project combined my experience as a fitness creator with data science skills like EDA, hypothesis testing, predictive modeling, and clustering. Analyzing my own Instagram data not only sharpened my technical abilities but also gave me clear, actionable strategies to prioritize shareable content over other engagement types. It reinforced the value of data-driven decision-making in shaping social media growth.